CareerMatch AI — Milestone II

Saron Lakew | Individual Final Project | Publish Your Agent

1. Public link and agent update

Public URL: NOT DEPLOYED YET — add the actual Streamlit link after publishing. This document is a working draft until the public URL and live app are checked.

I kept the same CareerMatch AI idea from Milestone I, but I connected three parts: a Profile Agent that reads a resume, a Job Search Agent that loads and filters saved job listings, and a Match Agent that compares skills and explains the results. The tool workflow has a four-call maximum so it cannot run forever. I also added optional Gemini explanations, but those still need a live API key test. All job listings are clearly labeled as fictional examples, not real openings.

Verified demo from my resume

```
CAREERMATCH AI — VERIFIED OFFLINE RUN
Actual PDF resume / Python tools / No Gemini API call

Status: complete     Stop: All planned tools completed
Skills identified: Python, SQL, Excel, Power BI, Tableau, Microsoft Copilot, Prompt Engineeri...

EXECUTION TRACE
1. Profile Agent | extract_profile
   Found 11 recognized skills; source: uploaded PDF (processed in memory).
2. Job Search Agent | search_jobs
   Loaded 4 fictional listings matching type/location.
3. Match Agent | rank_jobs
   Ranked 3 fictional listings; top overlap: 100.

TOP MATCHES
Business Intelligence Intern: 100% listed-skill overlap
Data Analyst Intern: 100% listed-skill overlap
Remote Data Intern: 67% listed-skill overlap

NOTE: These job listings are fictional demo records, not live openings.
```

The screenshot above is a rendering of the actual offline run trace, not a screenshot of a live Streamlit deployment. The complete machine-readable trace is included in the project folder.

2. Evaluation results

I ran 12 test cases three times each (36 total runs) against the offline agent. pass@1 means the first run passed. pass^3 means all three runs of the same case passed. The results below are for the Python workflow, not the Gemini API.

Test	Case	pass@1	pass^3
T01	Analytics matching	Pass	Pass
T02	BI requirements	Pass	Pass
T03	Software matching	Pass	Pass
T04	Full-time filter	Pass	Pass
T05	Remote preference	Pass	Pass
T06	Both job types	Pass	Pass
T07	No known skills	Pass	Pass
T08	No interest match	Pass	Pass
T09	Guard at step one	Pass	Pass
T10	Guard at step two	Pass	Pass
T11	Adversarial instruction in skills	Pass	Pass
T12	Adversarial huge input	Pass	Pass

Overall: 100% pass@1 and 100% pass^3 after the fix.

What failed and what I changed

During my first test, the unusually long skills input failed because the program silently cut it off. I changed it so the program rejects anything over 1,000 characters with a clear error message. After that, I reran the same 12 cases three times each and all passed. I also tested a prompt-injection style input asking for a secret key; the agent treated it as data and did not reveal a key. These tests do not guarantee that every possible input is safe.

3. Governance note

CareerMatch AI reads either a PDF resume or skills someone types in, along with their job type, career interest, and preferred location. The app only pulls out recognized skills and a degree label, and it compares those with a small CSV file of made-up job listings. The app does not save the uploaded resume to disk on purpose. Someone can download their own run results, and the test script saves evaluation results as JSON. If Gemini is turned on, it gets the extracted skills, degree label, and sample job details, not the full resume or contact information.

The human checkpoint is before the agents start: users have to check a box confirming that they understand these are sample jobs and that they will review the results themselves. There is also a separate switch for Gemini before anything is sent outside the app. The app never submits job applications, so the final choice stays with the user. I am responsible for the project and for fixing issues that come up. To limit costs and misuse, I cap the PDF size, manual input length, tool calls, and Gemini output length. For now, the biggest limitations are that the job data is only for a demo, the Streamlit website has not been published yet, and a real Gemini API call has not been tested.